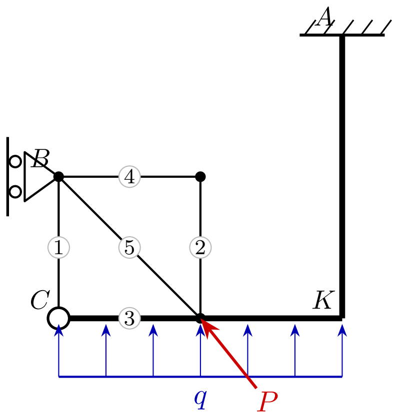
University of Natural Resources and Life Sciences, Vienna (BOKU)

**VU Mechanik – LAWI 100515 (PI)**

Exam **Group A** Date: 26.06.2026

## Model solution




### 1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies $k$, reactions $a$, interaction forces $z$):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \;\Rightarrow\; \text{statically determinate}$$

### 2. Support reactions and hinge forces

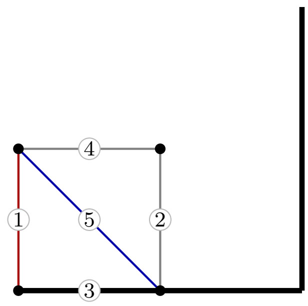
$A: \; R_x = -2, \; R_y = -25, \; M = 46.5 \quad B: \; R_x = 10, \; R_y = 0$

$C: \; H_x = 2, \; H_y = 10$

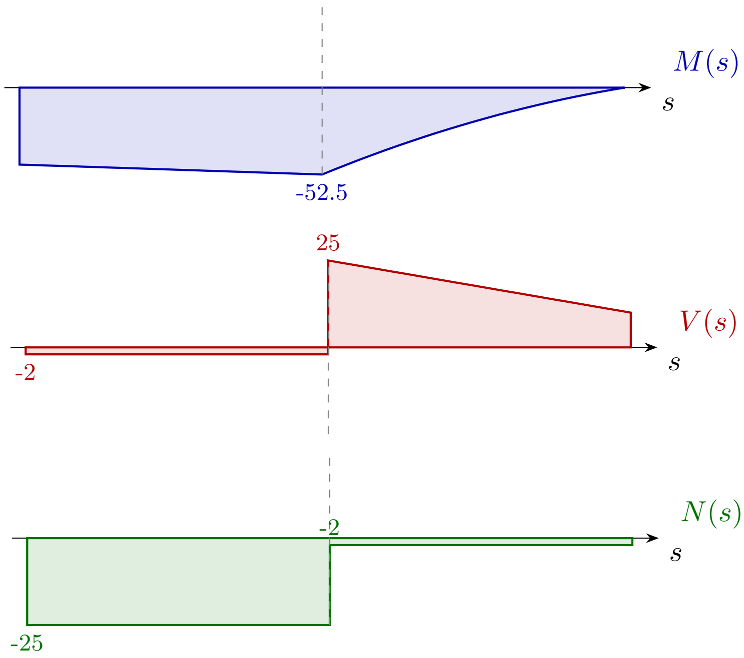
### 3. Truss member forces

| Member | Force $S$ [kN] | Type |
|---|---|---|
| 1 | 10 | tension |
| 2 | 0 | zero |
| 3 | 2 | tension |
| 4 | 0 | zero |
| 5 | -14.14 | compression |

red: tension (+) blue: compression (−)




### 4. Section-force diagrams of the beams



$M(s)$
$s$
-52.5
25
$V(s)$
$s$
-2
$N(s)$
-2
$s$
-25